

Supplementary material — A consistent continuum limit of mechanical relaxation models

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1 Derivation of continuum limit

Here we consider a one-dimensional chain of N springs mechanically relaxing. The positions of the i th spring boundaries at a given time t are given by $x_{i-1}(t)$ and $x_i(t)$ for $i = 1, \dots, N$. On the left we have $x_0(t)$ which is a fixed boundary (i.e. $dx_0/dt = 0$), and at the far right $x_N(t) = L(t)$ is a free boundary.

Newton's second law in an overdamped regime implies the position of the spring boundaries evolve according to

$$\eta \frac{dx_i}{dt} = f(x_{i+1} - x_i) - f(x_i - x_{i-1}), \quad i = 1, \dots, N-1, \quad (1)$$

where $f(\cdot)$ is the spring restoring force, and η is the drag coefficient. Here, we consider the Hookean force law

$$f(\ell_i) = k(\ell_i - a), \quad (2)$$

where $\ell_i = x_i - x_{i-1}$ is the length of the i th spring, k is the spring stiffness, and a is the resting spring length.

To derive the evolution equation for $x(t)$ and $L(t)$ as $m \rightarrow \infty$, we introduce a time-dependent parametrisation of the interface that tracks the evolving spring positions $\bar{x}_i(t)$, for constant parameter values u

$$X(u_i, t) = x_i(t), \quad (3)$$

where $u_i = i\Delta u$, $i = 0, \dots, M$, $M = mN$ are time-independent, evenly spaced coordinates in a finite domain $u \in [0, U)$. As $M \rightarrow \infty$, $\Delta u = U/M \rightarrow 0$ and $x_{i-1}(t) \sim x_i(t) \sim x_{i+1}(t)$. This parameterisation of the interface is such that

$$x_i(t) = \int_0^{u_i} du h(u, t), \quad h(u, t) = \frac{\partial X}{\partial u}. \quad (4)$$

Using Eq. (4) to express $x_{i\pm 1}(t)$ in terms of $u_i \pm \Delta u$ and expanding about as $\Delta u \rightarrow 0$, we obtain (omitting the time dependence of to simplify notation),

$$x_{i\pm 1} = x_i \pm h(u_i, t)\Delta u + \frac{\partial h}{\partial u}(u_i, t) \frac{\Delta u^2}{2} \pm \mathcal{O}(\Delta u^3). \quad (5)$$

We also introduce a continuous spring density $\rho(x_i(t), t) = 1/(x_i(t) - x_{i-1}(t))$ given at $x_i(t)$. By the parameterisation described in Eq. (4), spring density at $x_i(t)$ which corresponds to u_i is given by,

$$\rho(x_i(t), t) = \frac{1}{h(u_i, t)\Delta u}. \quad (6)$$

We first derive the evolution equation of the position of the internal spring boundaries, then we will derive the evolution of the boundary.

1.1 Derivation of continuum limit for internal nodes

For internal nodes we define the density of the springs at the midpoint $\bar{x}_i(t) = (x_{i-1}(t) + x_i(t))/2$. An alternate parameterisation for the midpoint is defined as $\bar{X}(u_i, t) = \bar{x}_i(t)$, with $\bar{x}_{i\pm 1}(t)$ and continuous density defined similarly as Eqs. (4)–(6). Substituting the Hookean restoring force from Eq. (2) into the discrete evolution equation for the internal node boundaries Eq. (1) gives,

$$\frac{dx_i}{dt} = \frac{k}{\eta} (x_{i+1} - 2x_i + x_{i-1}), \quad i = 1, \dots, N-1, \quad (7)$$

Rewriting Eq. (7) in terms of the midpoints of springs and substituting Eq. (5) gives

$$\frac{\partial \bar{X}}{\partial t}(u_i, t) = \frac{k}{\eta} \left[\Delta u^2 \frac{\partial h}{\partial u}(u_i, t) + \mathcal{O}(\Delta u^4) \right]. \quad (8)$$

Rearranging Eq. (6) to isolate $h(u_i, t)$, writing density ρ in terms of $\bar{X}(u, t)$ and differentiating $h(u, t)$ with respect to u yields

$$\frac{\partial h}{\partial u}(u_i, t) = -\frac{1}{\Delta u} \frac{1}{\rho(\bar{X}(u_i, t), t)^2} \frac{\partial \rho}{\partial \bar{X}}(\bar{X}(u_i, t), t) h(u_i, t), \quad (9)$$

since $\partial \bar{X} / \partial u = h(u, t)$. Substituting Eq. (6) and Eq. (9) into Eq. (8) gives,

$$\frac{\partial \bar{X}}{\partial t}(u_i, t) = -\frac{k}{\eta} \left[\frac{1}{\rho(\bar{X}(u_i, t), t)^3} \frac{\partial \rho}{\partial \bar{X}}(\bar{X}(u_i, t), t) + \mathcal{O}(\Delta u^4) \right]. \quad (10)$$

In the limit Eq. (10) becomes

$$\frac{\partial x}{\partial t} = -\frac{\alpha}{\rho^3} \frac{\partial \rho}{\partial x} + \mathcal{O}(\Delta u^4), \quad (11)$$

where $\alpha = k/\eta$. This result is the same as Eq. (21) from Baker et al. (2019).

1.2 Derivation of continuum limit for the free-boundary

At the free boundary $x_N(t)$, the evolution equation is given by

$$\eta \frac{dx_N}{dt} = -f(x_N - x_{N-1}). \quad (12)$$

Substituting Eq. (2) into Eq. (12) yields,

$$\frac{dx_N}{dt} = \frac{k}{\eta} (a - x_N + x_{N-1}). \quad (13)$$

Since the final boundary position is the length of the domain i.e. $x_N(t) = L(t)$, we can rewriting Eq. (13) in terms of spring midpoints and perform a Taylor expansion to give,

$$\frac{dL}{dt} = \frac{k}{\eta} \left(a - h(u_N, t) \Delta u + \frac{\partial h}{\partial u}(u_N, t) \frac{\Delta u^2}{2} - \mathcal{O}(\Delta u^3) \right). \quad (14)$$

Substituting Eq. (6) and Eq. (9) into Eq. (14) gives,

$$\frac{dL}{dt} = \frac{k}{\eta} \left(a - \frac{1}{\rho(L(t), t)} - \frac{1}{2} \frac{1}{\rho(L(t), t)^3} \frac{\partial \rho}{\partial \bar{X}}(L(t), t) + \mathcal{O}(\Delta u^3) \right). \quad (15)$$

Rearranging the equation above yields the following boundary condition in terms of spring density

$$\frac{dL}{dt} = \frac{k}{\eta} \left(a - \frac{1}{\rho(L(t), t)} \right) - \frac{1}{2} \frac{k}{\eta} \frac{1}{\rho(L(t), t)^3} \frac{\partial \rho}{\partial X}(L(t), t) + \mathcal{O}(\Delta u^3), \quad (16)$$

$$= \left[-\frac{1}{\eta} \tilde{f}(\rho) - \frac{D(\rho)}{2\rho} \frac{\partial \rho}{\partial x} + \mathcal{O}(\Delta u^3) \right]_{x=L(t)}, \quad (17)$$

where $\tilde{f}(\rho) = f(1/\rho)$ and $D(\rho) = k/\eta\rho^2$ is the diffusivity of the mechanical relaxation model with Hookean springs. This equation differs from the result shown in Baker et al. (2019).

2 Exact solution of M spring problem

Here, we investigate the initial-condition dependence of exact solutions of the evolution of the spring length $\ell_i(t) = x_i(t) - x_{i-1}(t)$ in a free boundary problem. In a generalised model of M springs we obtain the following linear system

$$\frac{d}{dt} \boldsymbol{\ell} = \frac{k}{\eta} \mathcal{A} \boldsymbol{\ell} + \frac{k}{\eta} \mathbf{y}, \quad (18)$$

where

$$\boldsymbol{\ell} = \begin{bmatrix} \ell_1(t) \\ \ell_2(t) \\ \vdots \\ \ell_{M-1}(t) \\ \ell_M(t) \end{bmatrix}, \quad \mathcal{A} = \begin{bmatrix} -1 & 1 & 0 & \cdots & 0 \\ 1 & -2 & \ddots & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & 1 & -2 & 1 \\ 0 & \cdots & 0 & 1 & -2 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 0 \\ \vdots \\ \vdots \\ 0 \\ a \end{bmatrix}.$$

By introducing a new time-dependent function $\phi_i(t)$ such that,

$$\phi_i(t) = \ell_i(t) - a, \quad (19)$$

and noting that $\phi'_i(t) = \ell'_i(t)$ we can rewrite the system as,

$$\boldsymbol{\phi}'(t) = \frac{k}{\eta} \mathcal{A} \boldsymbol{\phi}. \quad (20)$$

2.1 Solving for eigenvalues and eigenvectors

Expanding the rows of the eigenvector problem $\mathcal{A} \mathbf{v} = \lambda \mathbf{v}$ gives

$$v_2^{(p)} = (1 + \lambda_p) v_1^{(p)}, \quad j = 1, \quad (21)$$

$$v_{j+1}^{(p)} = (2 + \lambda_p) v_j^{(p)} - v_{j-1}^{(p)}, \quad j = 2, \dots, M-1, \quad (22)$$

$$v_{M-1}^{(p)} - (2 + \lambda_p) v_M^{(p)} = 0, \quad j = M. \quad (23)$$

To solve the recurrence in rows $2 \rightarrow M-1$, we assume that $v_j^{(p)} = r^j$ for some $r \neq 0$. Substituting this assumption into the recurrence relation gives

$$r^{j+1} = (2 + \lambda_p) r^j - r^{j-1}. \quad (24)$$

Dividing Eq. (24) by r^{j-1} gives,

$$r^2 = (2 + \lambda_p) r - 1. \quad (25)$$

The roots of this quadratic equation are then given by,

$$r_{1,2} = \frac{(2 + \lambda_p) \pm \sqrt{(2 + \lambda_p)^2 - 4}}{2}. \quad (26)$$

The general solution to the recurrence is then

$$v_j^{(p)} = Ar_1^j + Br_2^j. \quad (27)$$

Considering the discriminant of Eq. (26), since our eigenvalues are $-4 < \lambda_p < 0$ (shown numerically), this implies that $(2 + \lambda_p)^2 - 4 \leq 0$ and we have complex roots. Since we have complex roots and $-2 \leq \lambda_p + 2 \leq 2$, we make the substitution $\lambda_p + 2 = 2 \cos \theta_p$ for $\theta_p \in (0, \pi)$, rewriting Eq. (25) as

$$r^2 = (2 \cos \theta_p)r - 1, \quad (28)$$

and using the trig identity $\cos^2 \theta + \sin^2 \theta = 1$ we can rewrite $4 \cos^2 \theta_p - 4 = -4 \sin^2 \theta_p$ and Eq. (26) as

$$r = \frac{2 \cos \theta_p \pm \sqrt{-4 \sin^2 \theta_p}}{2} = \cos \theta_p \pm i \sin \theta_p = \exp(\pm i \theta_p). \quad (29)$$

Substituting Eq. (29) into Eq. (27) implies that the solution to the recurrence is,

$$v_j^{(p)} = A \exp(ij \theta_p) + B \exp(-ij \theta_p). \quad (30)$$

Expanding $\exp \pm i \theta_p$ using Euler's formula gives,

$$v_j^{(p)} = C \cos(j \theta_p) + D \sin(j \theta_p). \quad (31)$$

To solve for the coefficients C and D we introduce $v_0 = 1$ so that the recurrence holds everywhere, including at $j = 1$.

Proof. Letting $v_1^{(p)} = 1$, Eq. (21) becomes

$$v_2^{(p)} = (1 + \lambda_p)v_1, \quad (32)$$

$$= 2 \cos \theta_p - 1. \quad (33)$$

If we check the recurrence at $j = 1$ then we would get,

$$v_2^{(p)} = (2 + \lambda_p)v_1 - v_0, \quad (34)$$

$$= 2 \cos \theta_p - v_0, \quad (35)$$

which means that $v_0 = 1$ for the recurrence to hold at $j = 1$. □

Substituting $j = 0$ into Eq. (31) gives,

$$v_0^{(p)} = C \cos(0) + D \sin(0) \implies C = v_0 = 1. \quad (36)$$

At $j = 1$ we get,

$$v_1^{(p)} = \cos \theta_p + D \sin \theta_p \implies D = \frac{1 - \cos \theta_p}{\sin \theta_p}. \quad (37)$$

We can simplify the fraction in Eq. (37) using the half-angle trig identities $1 - \cos \theta = 2 \sin^2(\theta/2)$ and $\sin \theta = 2 \sin(\theta/2) \cos(\theta/2)$, to give

$$D = \frac{\sin(\theta_p/2)}{\cos(\theta_p/2)}. \quad (38)$$

Hence our solution to the recurrence is

$$v_j^{(p)} = \cos(j\theta_p) + \frac{\sin(\theta_p/2)}{\cos(\theta_p/2)} \sin(j\theta_p), \quad (39)$$

which can be simplified by multiplying the first term on the RHS of Eq. (39) by $\cos(\theta_p/2)/\cos(\theta_p/2)$ and using the trig identity $\cos(A - B) = \cos(A)\cos(B) + \sin(A)\sin(B)$ to give,

$$\begin{aligned} v_j^{(p)} &= \frac{\cos(j\theta_p)\cos(\theta_p/2) + \sin(j\theta_p)\sin(\theta_p/2)}{\cos(\theta_p/2)}, \\ &= \frac{\cos(j\theta_p - \theta_p/2)}{\cos(\theta_p/2)}, \\ &= \frac{\cos((j - 1/2)\theta_p)}{\cos(\theta_p/2)}. \end{aligned} \quad (40)$$

To solve for θ_p we consider Eq. (23). Substituting Eq. (40) into Eq. (23) gives,

$$\frac{\cos((M - 3/2)\theta_p)}{\cos(\theta_p/2)} = 2 \cos \theta_p \frac{\cos((M - 1/2)\theta_p)}{\cos(\theta_p/2)}. \quad (41)$$

Multiplying both sides by $\cos(\theta_p/2)$ and using the identity $2 \cos(A)\cos(B) = \cos(A + B) + \cos(A - B)$ in the RHS of Eq. (41), where $A = \theta_p$ and $B = (M - 1/2)\theta_p$ we obtain,

$$\cos((M + 1/2)\theta_p) = 0. \quad (42)$$

Since cosine is zero when the angle is an odd multiple of $\pi/2$, we can write,

$$\left(M + \frac{1}{2}\right) \theta_p = \frac{(2p - 1)\pi}{2}, \quad p = 1, 2, \dots, M \quad (43)$$

which gives,

$$\theta_p = \frac{(2p - 1)\pi}{2M + 1}. \quad (44)$$

Using the substitution $\lambda_p + 2 = 2 \cos \theta_p$ introduced above, the eigenvalues are given by,

$$\lambda_p = -2 + 2 \cos \theta_p, \quad \theta_p = \frac{(2p - 1)\pi}{2M + 1}. \quad (45)$$

2.2 Solving for coefficients

As in the two- and three-spring cases considered above, the coefficients are determined by the initial conditions. At $t = 0$ we can then write

$$\phi(0) = \sum_{k=1}^M c_k \mathbf{v}_k. \quad (46)$$

Since our matrix $\mathcal{A}$ is symmetric and has distinct eigenvalues, the dot product between any two different eigenvectors is zero, i.e. $\mathbf{v}_n \cdot \mathbf{v}_m = 0$ for $n \neq m$. Taking a dot product between both sides of Eq. (46) with $\mathbf{v}_p^T$ enables us to isolate each coefficient such that

$$\mathbf{v}_p^T \cdot \phi(0) = c_p (\mathbf{v}_p^T \cdot \mathbf{v}_p) = c_p \|\mathbf{v}_p\|^2. \quad (47)$$

Rearranging Eq. (47) gives

$$c_p = \frac{\mathbf{v}_p^T \cdot \phi(0)}{\|\mathbf{v}_p\|^2}. \quad (48)$$

Expanding $\mathbf{v}_p^T \cdot \boldsymbol{\phi}(0)$ gives,

$$\mathbf{v}_p^T \cdot \boldsymbol{\phi}(0) = \sum_{k=1}^M v_k^{(p)} \phi_k(0). \quad (49)$$

Substituting Eq. (40) into Eq. (49) and $\phi_k(0) = \ell_k(0) - a$ we obtain

$$\mathbf{v}_p^T \cdot \boldsymbol{\phi}(0) = \frac{1}{\cos(\theta_p/2)} \sum_{k=1}^M \cos\left((k-1/2)\theta_p\right) (\ell_k(0) - a). \quad (50)$$

To simplify notation we suppress the dependence on M of the eigenangle, writing $\theta_p = \theta_p(M)$. To calculate $\|\mathbf{v}_p\|^2$ we rewrite it as,

$$\|\mathbf{v}_p\|^2 = \sum_{k=1}^M \left(v_k^{(p)}\right)^2, \quad (51)$$

and substituting in Eq. (40) into the RHS of Eq. (51) gives,

$$\|\mathbf{v}_p\|^2 = \frac{1}{\cos^2(\theta_p/2)} \sum_{k=1}^M \cos^2\left((k-1/2)\theta_p\right). \quad (52)$$

Now we consider evaluating the sum in Eq. (52)

$$S = \sum_{k=1}^M \cos^2\left((k-1/2)\theta_p\right). \quad (53)$$

Using the double angle trig identity $\cos(2x) = 2\cos^2(x) - 1$, we can rewrite the cosine squared term as,

$$\cos^2\left((k-1/2)\theta_p\right) = \frac{1 + \cos\left((2k-1)\theta_p\right)}{2}. \quad (54)$$

Substituting this into Eq. (53) we obtain,

$$S = \frac{M}{2} + \frac{1}{2} \sum_{k=1}^M \cos\left((2k-1)\theta_p\right). \quad (55)$$

The sum of cosines can be simplified by multiplying it by $2\sin\theta_p$, using the trig identity $2\sin A \cos B = \sin(A+B) + \sin(A-B)$ and expanding the sum to give,

$$\begin{aligned} 2\sin\theta_p \sum_{k=1}^M \cos\left((2k-1)\theta_p\right) &= \sum_{k=1}^M \left[\sin\left(\theta_p + (2k-1)\theta_p\right) + \sin\left(\theta_p - (2k-1)\theta_p\right) \right], \\ &= \sum_{k=1}^M \left[\sin(2k\theta_p) - \sin\left((2k-2)\theta_p\right) \right], \\ &= [\sin(2\theta_p) - \sin(0)] + [\sin(4\theta_p) - \sin(2\theta_p)] + \dots \\ &\quad + [\sin(2M\theta_p) - \sin\left((2M-2)\theta_p\right)], \\ &= \sin(2M\theta_p) - \sin(0), \\ &= \sin(2M\theta_p). \end{aligned} \quad (56)$$

From Eq. (56) we obtain,

$$\sum_{k=1}^M \cos((2k-1)\theta_p) = \frac{\sin(2M\theta_p)}{2\sin\theta_p}. \quad (57)$$

Rearranging our definition for θ_p from Eq. (45) gives

$$2M\theta_p = (2p-1)\pi - \theta_p, \quad (58)$$

and since $\sin((2p-1)\pi - \theta_p) = \sin\theta_p$ for odd multiples of π Eq. (57) simplifies to

$$\sum_{k=1}^M \cos((2k-1)\theta_p) = \frac{\sin\theta_p}{2\sin\theta_p} = \frac{1}{2}. \quad (59)$$

Substituting this result into Eq. (55) we obtain,

$$S = \frac{2M+1}{4}, \quad (60)$$

and therefore,

$$\|\mathbf{v}_p\|^2 = \frac{2M+1}{4\cos^2(\theta_p/2)}. \quad (61)$$

Finally, substituting Eq. (50) and Eq. (61) into Eq. (48), we obtain the closed-form solution of c_p to be

$$c_p = \frac{4\cos(\theta_p/2)}{2M+1} \sum_{k=1}^M \cos((k-1/2)\theta_p) (\ell_k(0) - a). \quad (62)$$

2.3 Validation of closed-form solutions

The tables below compare the eigenvalues of small systems computed by hand with Eq. (45) and with the formula reported by Kouachi (2006),

$$\lambda_k = -2 + 2\cos\left(\frac{2k\pi}{2M+1}\right), \quad k = 1, 2, \dots, M, \quad (63)$$

for a matrix of the same structure. While the values in the table are rounded to three decimal places, the eigenvalues agreed to 16 decimal places between the hand calculations and Eq. (45). By contrast, Eq. (63) does not agree with either the hand calculations or Eq. (45) for the small systems considered here.

$M = 2$	Computed by hand	Eq. (45)	Kouachi (2006)
λ_1	$\frac{-3+\sqrt{5}}{2} \approx -0.382$	≈ -0.382	≈ -1.382
λ_2	$\frac{-3-\sqrt{5}}{2} \approx -2.618$	≈ -2.618	≈ -3.618
$M = 3$	Computed by hand	Eq. (45)	Kouachi (2006)
λ_1	≈ -0.198	≈ -0.198	≈ -0.753
λ_2	≈ -1.555	≈ -1.555	≈ -2.445
λ_3	≈ -3.247	≈ -3.247	≈ -3.802

Figure 1 compares the eigenvalues λ_p computed from Eq. (45) and Eq. (63), showing that the eigenvalues converge as the system size M becomes large. The convergence of the eigenvalues as $M \gg 1$ is expected, since both equations share the general form $\lambda_p = -2 + 2\cos\beta_p$ for their respective angles β_p , and β_p is $\mathcal{O}(1/M)$ in both cases.

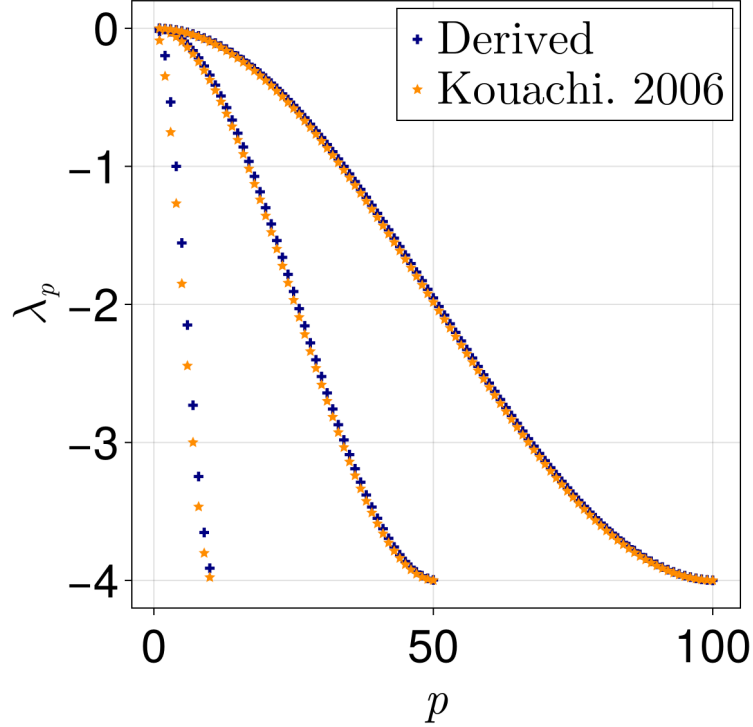


Figure 1: Comparison between computed eigenvalues using Eq. (45) and Eq. (63) for system size $M = 10, 50, 100$.

Similarly, in the tables below we compare the coefficients of a two- and three-spring system with the ones computed using Eq. (62). We validate Eq. (62) by considering springs in compression and under tension. For springs in compression, with $\ell_1(0) = \ell_2(0) = \ell_3(0) = 3$:

$M = 2$	Computed by hand	Eq. (62)
C_1	≈ -2.342	≈ -2.342
C_2	≈ 0.342	≈ 0.342
$M = 3$	Computed by hand	Eq. (62)
C_1	≈ -2.441	≈ -2.441
C_2	≈ 0.560	≈ 0.560
C_3	≈ -0.119	≈ -0.119

3 Rewriting in terms of cell lengths

Having obtained the exact solution of the M -spring problem, we now re-express it in terms of a fixed number of cells N , each composed of m springs, in preparation for the continuum limit $m \rightarrow \infty$. The exact solution to the evolution of the final spring length is given by

$$\ell_M(t) - a = \sum_{p=1}^M A_p \exp\left(-\frac{k}{\eta} \lambda_p t\right). \quad (64)$$

In the previous section we found the eigenvalues of $\mathcal{A}$ to be $-2 + 2 \cos \theta_p$, which are negative, so that each mode relaxes as $\exp((k/\eta)(-2 + 2 \cos \theta_p)t)$. It is therefore convenient to work with the (positive) decay rates

$$\lambda_p = 2 - 2 \cos \theta_p, \quad \theta_p = \frac{(2p-1)\pi}{2M+1}, \quad (65)$$

so that the relaxation is written with an explicit minus sign as $\exp(-(k/\eta)\lambda_p t)$. The generalised coefficient A_p is the product of the p th coefficient c_p and the M th component of the p th eigenvector, i.e. $A_p = c_p v_M^{(p)}$. From Eq. (40) we have

$$v_M^{(p)} = \frac{\cos((M-1/2)\theta_p)}{\cos(\theta_p/2)}, \quad (66)$$

and, from Eq. (62), the p th coefficient is given by

$$c_p = \frac{4 \cos(\theta_p/2)}{2M+1} \sum_{k=1}^M \cos\left((k-1/2)\theta_p\right) (\ell_k^*(0) - a). \quad (67)$$

Combining these, we obtain the generalised coefficient as

$$A_p = \frac{\cos((M-1/2)\theta_p)}{\cos(\theta_p/2)} \frac{4 \cos(\theta_p/2)}{2M+1} \sum_{k=1}^M \cos\left((k-1/2)\theta_p\right) (\ell_k^*(0) - a). \quad (68)$$

To convert the system of M springs into a finite number of cells N , each containing m springs, we use the following parameter transformations

$$M = Nm, \quad k = k^*m, \quad \eta = \frac{\eta^*}{m}, \quad a = \frac{a^*}{m}, \quad (69)$$

where k^* , η^* , and a^* are cell-related quantities. We assume that each cell contains m equally spaced springs, so that the cell length is $\ell_i^* = m\ell_j$, where $i = 1, 2, \dots, N$ indexes the cell and j indexes a spring within the i th cell. Applying Eq. (69) to Eqs. (64), (68) and (65), we obtain

$$m(\ell_N^*(t) - a^*) = \sum_{p=1}^{Nm} A_p(m) \exp\left(-\frac{k^*}{\eta^*} m^2 \lambda_p(m) t\right), \quad (70)$$

such that

$$\lambda_p(m) = 2 - 2 \cos(\theta_p(m)), \quad \theta_p(m) = \frac{(2p-1)\pi}{2Nm+1}, \quad (71)$$

$$A_p(m) = \frac{\cos((Nm-1/2)\theta_p(m))}{\cos(\theta_p(m)/2)} \frac{4 \cos(\theta_p(m)/2)}{2Nm+1} \sum_{k=1}^{Nm} m \cos\left((k-1/2)\theta_p(m)\right) (\ell_k^*(0) - a^*). \quad (72)$$

3.1 Taking the limit as $m \rightarrow \infty$

Here we consider the limit as $m \rightarrow \infty$. As m becomes large, the eigenangles $\theta_p(m)$ can be approximated as

$$\frac{(2p-1)\pi}{2Nm+1} \approx \frac{(2p-1)\pi}{2Nm}. \quad (73)$$

For a finite value of p , $\theta_p \rightarrow 0$ as $m \rightarrow \infty$, which allows us to analyse $\lambda_p(m)$ via Taylor expansion in the limit. Taylor expanding the cosine in the eigenvalues about 0 with $\Delta\theta = \theta_p$ as $m \rightarrow \infty$ gives

$$\begin{aligned} \lambda_p(m) &= 2 - 2 \left(1 - \frac{\theta_p^2}{2} + \mathcal{O}(\theta_p^4)\right), \\ &\approx \frac{(2p-1)^2 \pi^2}{4N^2 m^2}. \end{aligned} \quad (74)$$

For finite constant values of p (slow eigenmodes),

$$m^2 \lambda_p(m) \rightarrow \frac{(2p-1)^2 \pi^2}{4N^2}, \quad \text{as } m \rightarrow \infty, \quad (75)$$

which implies that the entire exponent converges to

$$-\frac{k^*}{\eta^*} m^2 \lambda_p(m) t \rightarrow -\frac{k^*(2p-1)^2 \pi^2 t}{4N^2 \eta^*}, \quad \text{as } m \rightarrow \infty. \quad (76)$$

3.1.1 Simplifying $A_p(m)$

The coefficient A_p comprises three factors. The first factor can be simplified by rewriting the argument of the cosine in the numerator,

$$\begin{aligned} (Nm - 1/2)\theta_p(m) &= \frac{2Nm - 1}{2} \left(\frac{(2p-1)\pi}{2Nm+1} \right), \\ &= \frac{(2p-1)\pi}{2} \left(\frac{2Nm-1}{2Nm+1} \right), \\ &= \frac{(2p-1)\pi}{2} \left(\frac{(2Nm+1)-2}{2Nm+1} \right), \\ &= \frac{(2p-1)\pi}{2} \left(1 - \frac{2}{2Nm+1} \right), \\ &= \frac{(2p-1)\pi}{2} - \theta_p(m). \end{aligned} \quad (77)$$

Expanding the cosine in the numerator with the trig identity $\cos(A-B) = \cos(A)\cos(B) + \sin(A)\sin(B)$ gives,

$$\cos\left(\frac{(2p-1)\pi}{2} - \theta_p(m)\right) = \underbrace{\cos\left(\frac{(2p-1)\pi}{2}\right)}_{=0} \cos(\theta_p(m)) + \underbrace{\sin\left(\frac{(2p-1)\pi}{2}\right)}_{=(-1)^{p+1}} \sin(\theta_p(m)). \quad (78)$$

Since p is an integer, the first cosine is zero and the first sine takes the value of ± 1 . Using the double angle formula $\sin(\theta_p(m)) = 2 \cos(\theta_p(m)/2) \sin(\theta_p(m)/2)$, the first factor becomes

$$\frac{\cos((Nm - 1/2)\theta_p(m))}{\cos(\theta_p(m)/2)} = (-1)^{p+1} \cdot 2 \sin(\theta_p(m)/2). \quad (79)$$

To simplify the second factor we consider the limit as m becomes large. In the limit as $m \rightarrow \infty$, $\theta_p(m) \rightarrow 0$ for finite values of p , and taking the leading order of a Taylor expansion of $\cos(\theta_p(m)/2) = 1 + \mathcal{O}(1/m^2)$ we obtain

$$\frac{4 \cos(\theta_p(m)/2)}{2Nm+1} \approx \frac{4}{2Nm} \approx \frac{2}{Nm}, \quad \text{as } m \rightarrow \infty. \quad (80)$$

The third factor, which is the sum, can be treated in two cases: (i) when all initial conditions are equal, $\ell_k(0) = \ell(0)$ for all $k = 1, 2, \dots, Nm$, and (ii) when the initial conditions $\ell_k(0)$ vary. In this document we consider only the first case, in which the springs are initially uniformly distributed. When $\ell_k(0) = \ell(0)$ for all $k = 1, 2, \dots, Nm$, the sum can be rewritten as

$$\sum_{k=1}^{Nm} \cos\left((k-1/2)\theta_p(m)\right) m(\ell_k^*(0) - a^*) = m(\ell^*(0) - a^*) \sum_{k=1}^{Nm} \cos\left((k-1/2)\theta_p(m)\right). \quad (81)$$

From our earlier derivation of the closed-form solution (Eq. (57)) we obtained the result

$$\sum_{k=1}^M \cos((2k-1)\alpha) = \frac{\sin(2M\alpha)}{2\sin\alpha}. \quad (82)$$

If we let $\alpha = \theta_p(m)/2$ and rewrite $M = Nm$ we obtain

$$\sum_{k=1}^{Nm} \cos((k-1/2)\theta_p(m)) = \frac{\sin(Nm\theta_p(m))}{2\sin(\theta_p(m)/2)}. \quad (83)$$

Recalling our definition of $\theta_p(m)$ from Eq. (71) we get,

$$Nm\theta_p(m) = \frac{(2p-1)\pi}{2} - \frac{\theta_p(m)}{2}, \quad (84)$$

and expanding the sine in the denominator using the identity $\sin(A-B) = \sin(A)\cos(B) - \cos(A)\sin(B)$, and substituting Eqs. (83) and (84) into Eq. (81), yields

$$m(\ell^*(0) - a^*) \sum_{k=1}^{Nm} \cos((k-1/2)\theta_p(m)) = m(\ell^*(0) - a^*) \frac{(-1)^{p+1} \cos(\theta_p(m)/2)}{2\sin(\theta_p(m)/2)}. \quad (85)$$

Combining Eqs. (79), (80) and (85) gives,

$$A_p(m) \sim \frac{4\cos^2(\theta_p(m)/2)}{2Nm+1} m(\ell^*(0) - a^*). \quad (86)$$

Bounds of $A_p(m)$ in the limit as $m \rightarrow \infty$: Before splitting the sum, we must verify that A_p depends on m algebraically rather than exponentially. To verify this, we consider the bounds of $|A_p|$. From Eq. (71) we know that $0 < \theta_p(m) < \pi$ for $p = 1, \dots, Nm$ and all values of m , therefore

$$|\cos^2(\theta_p(m)/2)| \leq 1 \quad \forall \theta_p(m). \quad (87)$$

We can also write that

$$\frac{4m}{2Nm+1} \leq \frac{4m}{2Nm} = \frac{2}{N}. \quad (88)$$

Combining Eqs. (87) and (88) we can write,

$$|A_p| \leq \frac{2}{N} |\ell^*(0) - a^*|, \quad (89)$$

for all values of p and m . This result indicates that the bound on the coefficient A_p is independent of both p and m .

3.2 Splitting up the summation

Suppose we can write the right-hand side of Eq. (70) as

$$\sum_{p=1}^{Nm} A_p \exp\left(-\frac{k^* m^2 \lambda_p(m) t}{\eta^*}\right) = \sum_{p=1}^{P(m)} A_p \exp\left(-\frac{k^* m^2 \lambda_p(m) t}{\eta^*}\right) + \sum_{p>P(m)}^{Nm} A_p \exp\left(-\frac{k^* m^2 \lambda_p(m) t}{\eta^*}\right), \quad (90)$$

where the exponents in the first sum are $\mathcal{O}(1)$ and those in the second sum are $\mathcal{O}(m^\nu)$ for any $\nu > 0$, whose contributions are negligible as $m \rightarrow \infty$. To determine $P(m)$, we require

$$\lambda_p(m) m^2 \geq c m^\nu. \quad (91)$$

Using Eq. (71) we can solve this inequality to obtain

$$p \geq \frac{1}{2} + \frac{Nm + 1/2}{\pi} \arccos \left(1 - \frac{c}{2m^{2-\nu}} \right). \quad (92)$$

Letting $\epsilon = 1/m \rightarrow 0^+$ as $m \rightarrow \infty$ and using L'Hôpital's rule, we obtain the approximation of the lower bound to be

$$P(m) \sim \frac{N\sqrt{c}}{\pi} m^{\nu/2}, \quad m \rightarrow \infty. \quad (93)$$

Since $\nu > 0$ and can be arbitrarily small, the lower bound $P(m) \rightarrow \infty$ much slower than $m \rightarrow \infty$. These results indicate that $P(m) \ll m \implies p \ll m$, and therefore

$$2 - 2\cos(\theta_p(m)) \sim \frac{(p - 1/2)^2 \pi^2}{(Nm + 1/2)^2} \implies \lambda_p(m) m^2 \sim \frac{(p - 1/2)^2 \pi^2}{N^2}. \quad (94)$$

Given that our analysis indicated that the contributions of the terms in the second sum of Eq. (90) are negligible, using Eq. (94) we can rewrite the first sum in Eq. (90) as

$$\begin{aligned} \lim_{m \rightarrow \infty} \sum_{p=1}^{Nm} A_p \exp \left(-\frac{k^* m^2 \lambda_p(m) t}{\eta^*} \right) &= \lim_{m \rightarrow \infty} \sum_{p=1}^{P(m)} A_p(m) \exp \left(-\frac{k^* (p - 1/2)^2 \pi^2 t}{\eta^* N^2} \right), \\ &= \sum_{p=1}^{\infty} \tilde{A} \exp \left(-\frac{k^* (p - 1/2)^2 \pi^2 t}{\eta^* N^2} \right), \end{aligned} \quad (95)$$

where $\tilde{A} = 2/N(\ell^*(0) - a^*)$

4 Converting the outer sum into an integral

To simplify notation before converting the sum to an integral using the Euler–Maclaurin formula, we rewrite the time scale as

$$\tau = \frac{k^* \pi^2 t}{\eta^* N^2}. \quad (96)$$

The Euler–Maclaurin formula enables us to convert the infinite sum to an integral via

$$\sum_{i=a}^b f(i) = \int_a^b dx f(x) + \frac{f(a) + f(b)}{2} + \sum_{k=1}^{\frac{n}{2}} \frac{B_{2k}}{(2k)!} (f^{(2k-1)}(b) - f^{(2k-1)}(a)) + R_n, \quad (97)$$

where $f(x)$ is the function of the parameter which is summed over on the right-hand side, B_k is the k th Bernoulli number, and R_n is an error term which depends on a , b , n and f . In the case of the sum presented in Eq. (95), the bounds are given by $a = 1$ and $b = \infty$, and the function f is

$$f(p) = \exp \left(-(p - 1/2)^2 \tau \right). \quad (98)$$

Using the substitution $x = (p - 1/2)\sqrt{\tau}$, the function in Eq. (98) in terms of x then becomes the Gaussian,

$$f(x) = \exp \left(-x^2 \right). \quad (99)$$

At the bound $b = \infty$, $f(b) \rightarrow 0^+$, leading to

$$\begin{aligned} \tilde{A} \sum_{p=1}^{\infty} \exp \left(-(p - 1/2)^2 \tau \right) &= \tilde{A} \left(\frac{1}{\sqrt{\tau}} \int_{\sqrt{\tau}/2}^{\infty} dx \exp \left(-x^2 \right) + \frac{1}{2} \exp \left(-\frac{\tau}{4} \right) + (\text{Bernoulli corrections}) \right), \\ &= \frac{\tilde{A}}{2} \left(\sqrt{\frac{\pi}{\tau}} \operatorname{erfc} \left(\frac{\sqrt{\tau}}{2} \right) + \exp \left(-\frac{\tau}{4} \right) + 2(\text{Bernoulli corrections}) \right), \end{aligned} \quad (100)$$

For sufficiently small values of τ (i.e. as $\tau \rightarrow 0^+$), $\operatorname{erfc}(\sqrt{\tau}/2) = 1 - \mathcal{O}(\sqrt{\tau})$ and $\exp(-\tau/4) \rightarrow 1$. Considering only the leading-order terms, at early times we obtain

$$\sum_{p=1}^{\infty} \tilde{A} \exp(-(p-1/2)^2\tau) \sim \frac{\tilde{A}}{2} \sqrt{\frac{\pi}{\tau}} \quad \text{as } \tau \rightarrow 0^+. \quad (101)$$

Note that between Eq. (100) and Eq. (101) we removed the Bernoulli corrections and R_n error term. In the following subsection we show that the Bernoulli corrections are negligible and tend to zero as $\tau \rightarrow 0^+$. The error term R_n is strictly smaller than or equal to the Bernoulli corrections, and by extension is also negligible.

4.1 Negligible Bernoulli corrections

The Bernoulli corrections in Eq. (97) are negligible for the following reason. Since f and all of its derivatives vanish as $x \rightarrow \infty$, only the boundary at $x = 1$ contributes. To simplify the working we will let $u = p - 1/2$ and hold τ as a constant. The first two odd derivatives $(2k-1)$ are

$$f'(u) = -2\tau u \exp(-\tau u^2), \quad f''' = (12\tau^2 u - 8\tau^3 u^3) \exp(-\tau u^2). \quad (102)$$

When $p = 1$ we have $u = 1/2$. Evaluating these derivatives at $u = 1/2$ gives

$$f'(1/2) = -\tau \exp\left(-\frac{\tau}{4}\right), \quad f'''(1/2) = (6\tau^2 - \tau^3) \exp\left(-\frac{\tau}{4}\right). \quad (103)$$

Given that the first two Bernoulli terms are $B_2 = 1/6$ and $B_4 = -1/30$, the leader order corrections for small τ are therefore,

$$(\text{Bernoulli corrections}) \approx -\frac{B_2}{2!} f'(1) - \frac{B_4}{4!} f'''(1) \approx \left(\frac{\tau}{12} + \frac{\tau^2}{120} + \dots \right) \exp\left(-\frac{\tau}{4}\right) \approx \frac{\tau}{12} - \frac{\tau^2}{80} + \mathcal{O}(\tau^3). \quad (104)$$

Each of the following terms carry an additional power of τ , so they vanish faster than the leading term in Eq. (101) as $\tau \rightarrow 0^+$.